CHAPTER 3 – The Simplex Method

In LP the optimum solution is always associated with a corner point of the feasible region. Simplex method looks at the corner points (extreme points) in a systematic manner. It stops when no further improvement is possible.

Simplex method is best explained if the LP is in equation (standard) form.

In Equation (standard) form:

i. All constraints are equations with non-negative RHS.
ii. All variables are non-negative.

- To convert (≤) into (=) add a non-negative slack variable (s) to LHS;

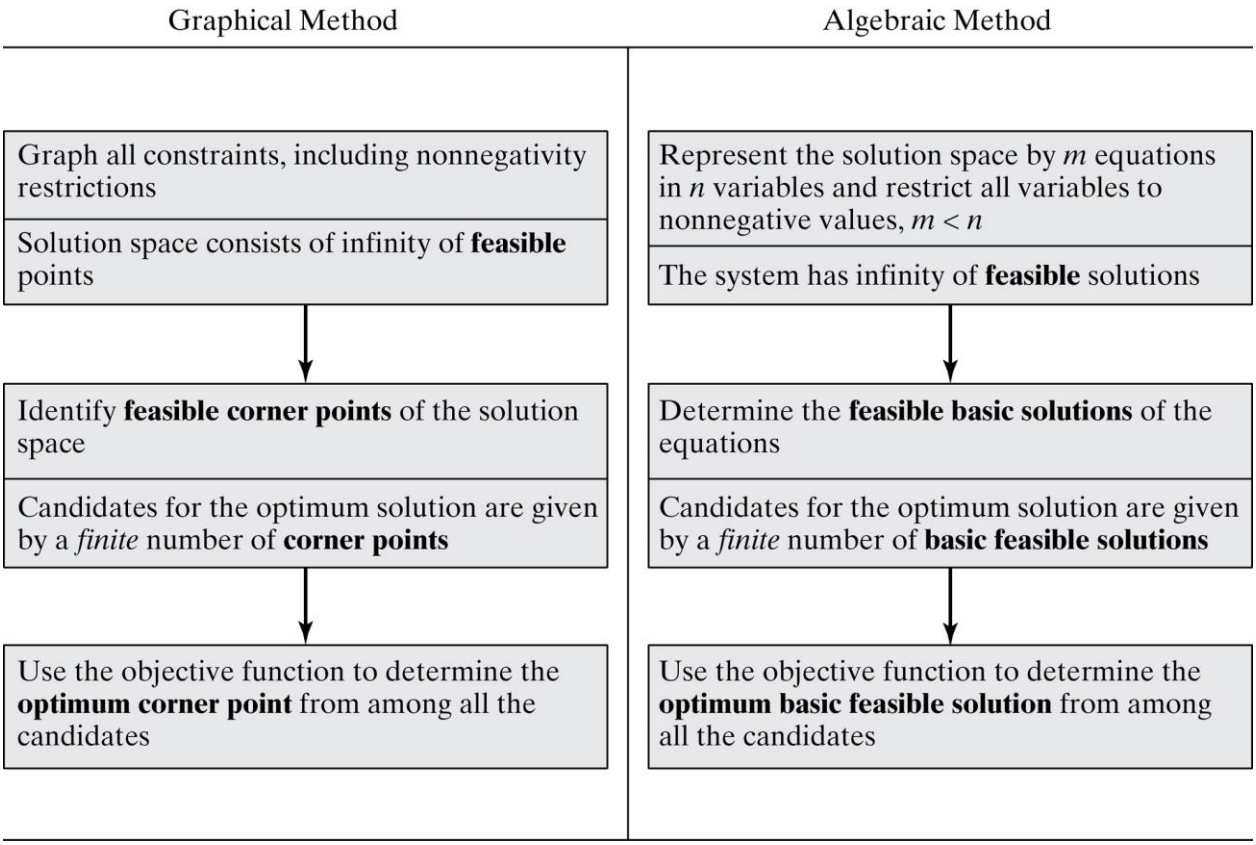
- To convert (≥) into (=) subtract a non-negative surplus variable (e);

- If RHS is negative, multiply both sides by (-1);

- Unrestricted variables

- LP solution can never be yield positive values for x_3^+ *and* x_3^- simultaneously
- If x_3 is in the objective function or other constraints it should be replaced by $\left(x_3^+ - x_3^-\right)$.

Transition from Graphical to Algebraic Solution



In the algebraic solution space (defined by m x n equations, m<n), **basic solutions** correspond to the corner points in the graphical solution space.
How can I extract the corner points algebraically?
They are determined by setting n-m variables equal to zero and solving m equations for the remaining m variables.

Corner points ←→ basic feasible solutions

m equations
n variables

In LP usually m<n
o/w if m=n → unique solution
m>n → at least m-n equations must be redundant

m<n

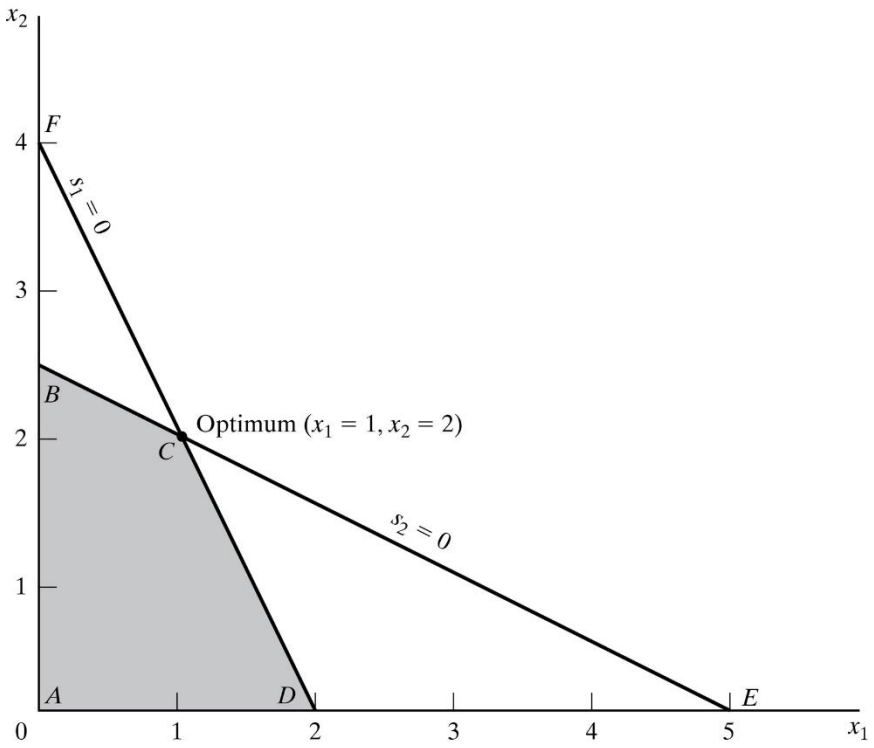
Set (n-m) variables to zero then solve the $m \times m$ equation system.

- Such a solution corresponds to a corner point.

Example:

Max $z = 2x_1 + 3x_2$
s.t.

$2x_1 + x_2 \le 4$
$x_1 + 2x_2 \le 5$ ➔
$x_1, x_2 \ge 0$



max $z = 2x_1 + 3x_2$

s.t.

$$2x_1 + x_2 + s_1 = 4$$
$$x_1 + 2x_2 + s_2 = 5$$
$$x_1, x_2, s_1, s_2 \geq 0$$

set for example $x_1 = x_2 = 0 \rightarrow s_1=4, s_2=5$ } point A

Let us obtain the other corner points:

Note that E&F are infeasible!!!

- How many combinations are there for *m* equations and *n* variables?
- Which (n-m) variables should be set to zero?

(n-m) variables set to zero → non-basic variables
use m variables → basic variables →basic solution

If we are to investigate all basic solutions, it could be a computational burden.

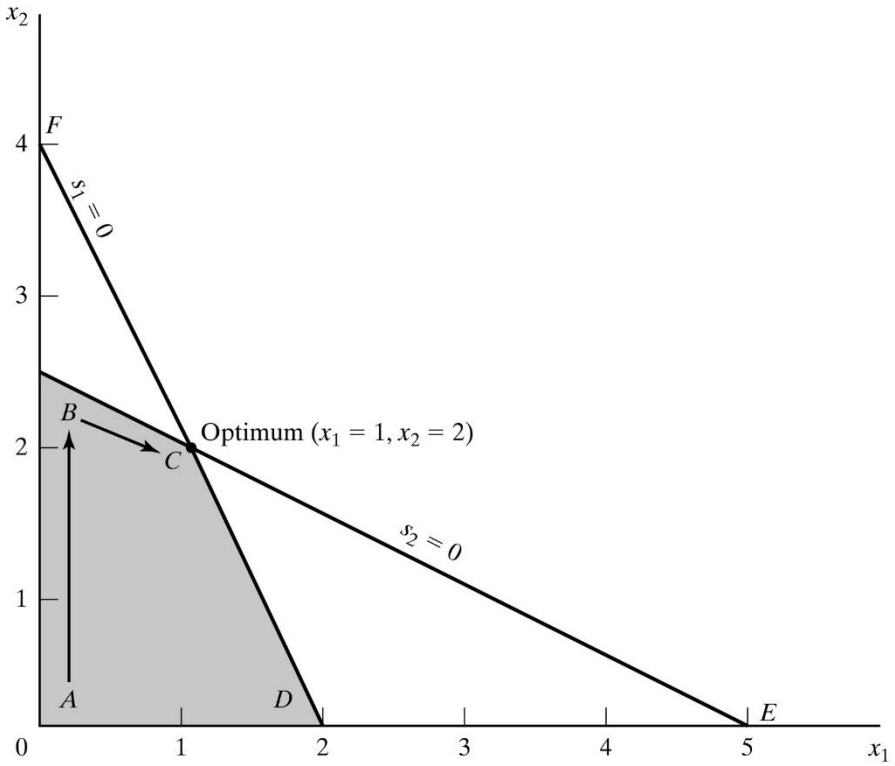
$c\binom{20}{10} = 184{,}756$ sets of 10x10 equation systems should be solved.

20x10 is practically a small LP.

- Simplex method performs an intelligent search procedure among the extreme points.

Iterative Nature of the Simplex Method

- Simplex method usually starts at origin.
- At each iteration 1 basic variable changes
- The non-basic variable which improves the objective function most is selected to be basic.



A → B → C : Simplex path

At A $x_1 = x_2 = 0$

max $z = 2x_1 + 3x_2$

x_2 has a larger coefficient

→ move to B

At point B, the simplex method will increase the value of x_1 to reach the corner point C, the optimum.

The path of simplex algorithm always connects corner points.

It always moves alongside the edges of the solution space, i.e. it does not cut across the solution space.

The Simplex Method

Remember the Reddy Mikks example:

Max $z = 5x_1 + 4x_2$

s.t.

$$\begin{aligned} 6x_1 + 4x_2 &<= 24 \quad (M1) \\ x_1 + 2x_2 &<= 6 \quad (M2) \\ -x_1 + x_2 &<= 1 \quad (demand) \\ x_2 &<= 2 \quad (demand) \end{aligned}$$ →

Simplex Tableau

The algorithm starts at the origin, i.e. $(x_1, x_2)=(0,0)$

- Non-basic → (x_1, x_2)
- Basic → (s_1, s_2, s_3, s_4)

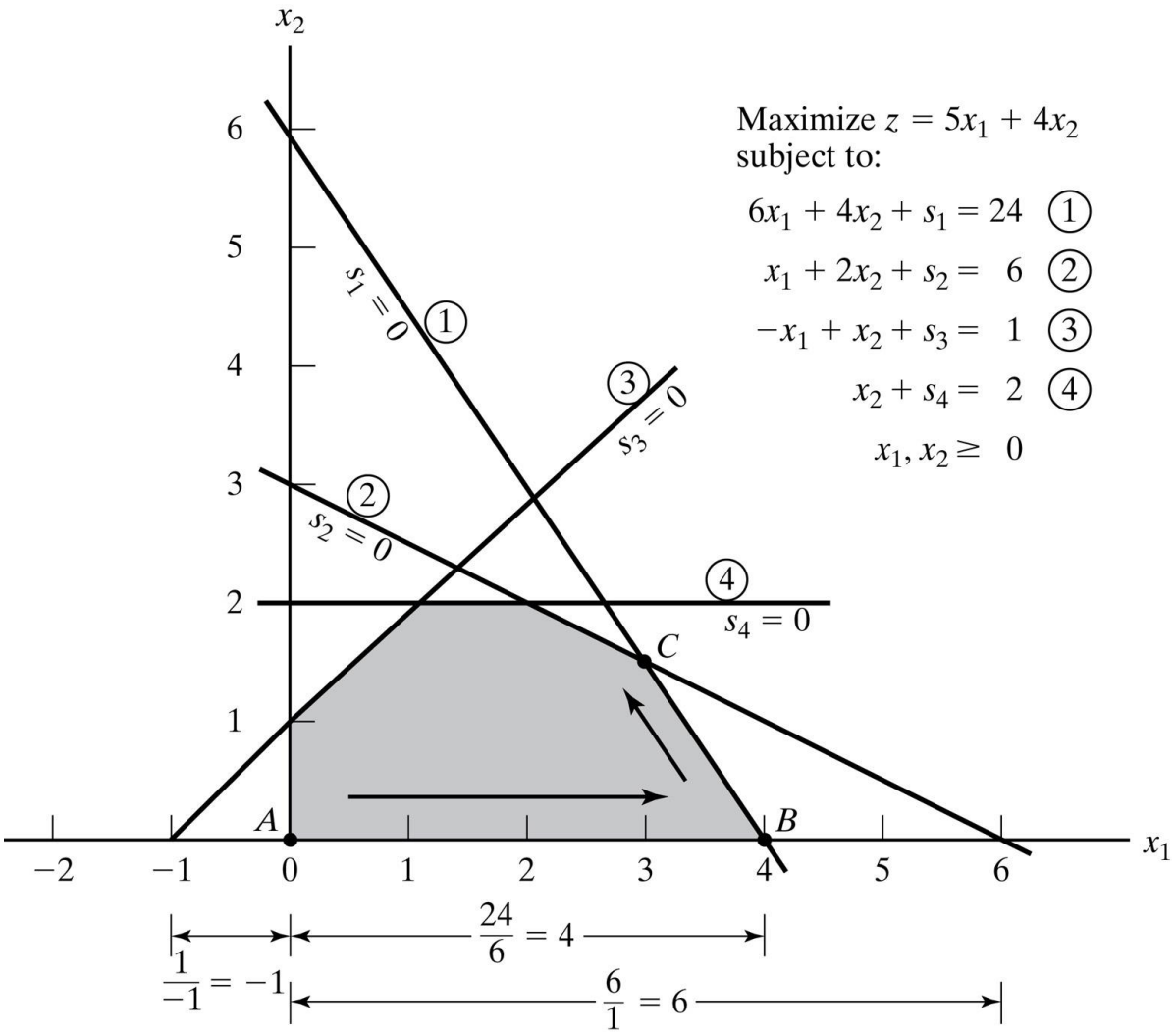
$z = 0$
$s_1 = 24$
$s_2 = 6$
$s_3 = 1$
$s_4 = 2$

} Due to the identity matrix

$z = 5x_1 + 4x_2$ or $z - 5x_1 - 4x_2 = 0$ → x_1 should enter the basis (entering variable)

↑ Most positive ↑ Most negative

- The z equation shows that the solution can be improved by improving the value of x_1 or x_2 above zero. x_1 is selected because it has the most positive objective coefficient.
- This rule defines the so-called **simplex optimality condition**.
- If one is entering the basic variable set, one must leave (because # of non-basic variable must always be n-m)



Maximize $z = 5x_1 + 4x_2$
subject to:

$$6x_1 + 4x_2 + s_1 = 24 \quad (1)$$
$$x_1 + 2x_2 + s_2 = 6 \quad (2)$$
$$-x_1 + x_2 + s_3 = 1 \quad (3)$$
$$x_2 + s_4 = 2 \quad (4)$$
$$x_1, x_2 \geq 0$$

To determine the leaving basic variable, apply minimum ratio (MR) test: (this assures feasibility)

when x_1 is entering the basis, we compute the intercepts of all the constraints with the non-negative direction of x_1-axis: These intercepts are the ratios.

MR appeared in s_1-row
→ s_1 is the leaving variable
→ x_1=4 (the MR) (point B)

non-basic → (s_1, x_2)
basic → (x_1, s_2, s_3, s_4)
} Point B

To obtain the new tableau apply the Gauss-Jordan row operations:

1. Pivot row
   New pivot row = current pivot row / pivot element
2. All other rows including z
   New row = current row – (its pivot col. Coef.) x (new pivot row)

   New pivot row = current row / 6
   New z-row = current z-row – (-5) x new pivot row
   New s_2-row = current s_2-row – (1) x new pivot row
   New s_3-row = current s_3-row – (-1) x new pivot row
   New s_4-row = current s_4-row – (0) x new pivot row

$z = \frac{2}{3}x_2 - \frac{5}{6}s_1 + 20$ Increasing x_2 is beneficial

New pivot row = current row / 1.5
New z-row = current z-row – (-2/3) x new pivot row
New x_1-row = current x_1-row – (2/3) x new pivot row
New s_3-row = current s_3-row – (5/3) x new pivot row
New s_4-row = current s_4-row – (1) x new pivot row

Since z-row coefficient of all non-basic variables are non-negative, the current tableau is optimal with:

$x_1 = 3$
$x_2 = 3/2$
$s_3 = 5/2$
$s_4 = 1/2$
$z = 21 = 5x_1 + 4x_2 = 5 \times 3 + 4 \times \frac{3}{2} = 21$

$s_1=0$ → M_1 is used completely → M_1 is scarce
$s_2=0$ → M_2 is used completely → M_2 is scarce

if $s_i > 0$ → M_i is abundant

In a minimization problem everything is the same except that the entering variable is determined as the one having most positive coefficient in z-row.

Simplex Method Summary

Optimality Condition: In a max (min) problem the non-basic variable having the most negative (positive) coefficient in the z-row is the entering variable.

Feasibility Condition: For both max and min problems the basic variable having the smallest non-negative ratio is the leaving variable.

Step 0. Determine a starting basic feasible solution.
Step 1. Select the entering variable using the optimality condition.
Stop if there is no entering variable; last solution is optimal.
Step 2. Select a leaving variable using the feasibility condition.
Step 3. Determine the new basic solution by Gauss-Jordan row operations. Go to step 1.

Example:

Example:

Artificial Starting Solutions

- It is important to have an identity matrix to start the simplex iterations.
- If all constraints are ≤ type, then the identity matrix is readily available using the slack variables.
- In case there are (=) and (≥) constraints starting is not that easy.
- At the initial phases artificial variables are used.
- There are two common methods: M-method (big M method)
  Two-phase method

LPs in which all the constraints are <= with nonnegative RHSs offer a convenient all-slack starting basic feasible solution (all non-slacks are set to zero if you remember the example i.e. $(x_1,x_2)=(0,0)$.
But models having >= and/or = do not.
In these models the origin point is not guaranteed to be a feasible point to start.

In this case, we use artificial variables that play the role of slacks in the first phase. Then, they are disposed and we continue to Phase II if Phase I ends successfully.
We add an artificial variable to all = and >= equations as we illustrate in the following example.

Two-Phase Method

Phase 1: Put the problem in equation form
Add the necessary artificial variables
Find a basic solution which minimizes the sum of artificial variables
→ If the sum is positive stop, LP is infeasible
→o/w proceed to phase 2

Phase 2: Use the feasible solution from phase 1 as the starting basic feasible solution for the original problem.

Example:

$$\begin{aligned} \min\ & z = 4x_1 + x_2 \\ s.t.\ & 3x_1 + x_2 = 3 \\ & 4x_1 + 3x_2 \geq 6 \\ & x_1 + 2x_2 \leq 4 \\ & x_1, x_2 \geq 0 \end{aligned} \quad \rightarrow \quad \begin{aligned} \min\ & z = 4x_1 + x_2 \\ s.t.\ & 3x_1 + x_2 = 3 \\ & 4x_1 + 3x_2 - e_1 = 6 \\ & x_1 + 2x_2 + s_1 = 4 \\ & x_1, x_2, e_1, s_1 \geq 0 \end{aligned}$$

$$\begin{aligned} \min\ & r = R_1 + R_2 \\ s.t.\ & 3x_1 + x_2 + R_1 = 3 \\ & 4x_1 + 3x_2 - e_1 + R_2 = 6 \\ & x_1 + 2x_2 + s_1 = 4 \\ & x_1, x_2, x_3, x_4, R_1, R_2 \geq 0 \end{aligned}$$

Basic	x1	x2	e1	R1	R2	s1	solution
r	0	0	0	-1	-1	0	0
R1	3	1	0	1	0	0	3
R2	4	3	-1	0	1	0	6
s1	1	2	0	0	0	1	4

Basic	x1	x2	e1	R1	R2	s1	solution
r	0	0	0	-1	-1	0	0
x1	1	0	1/5	3/5	-1/5	0	3/5
x2	0	1	-3/5	-4/5	3/5	0	6/5
s1	0	0	1	1	-1	1	1

Phase-1 optimal $r^*=0$ → get rid of R_1 and R_2 columns.

Now,

$$\begin{aligned} \min \quad & z = 4x_1 + x_2 \\ s.t. \quad & x_1 + \frac{1}{5}x_3 = \frac{3}{5} \\ & x_2 - \frac{3}{5}x_3 = \frac{6}{5} \\ & x_3 + x_4 = 1 \end{aligned}$$

Basic	x1	x2	e1	s1	Solution
z	-4	-1	0	0	0
x1	1	0	1/5	0	3/5
x2	0	1	-3/5	0	6/5
s1	0	0	1	1	1
z	0	0	1/5	0	18/5
x1	1	0	1/5	0	3/5
x2	0	1	-3/5	0	6/5
s1	0	0	1	1	1
z	0	0	0	-1/5	17/5
x1	1	0	0	-1/5	2/5
x2	0	1	0	3/5	9/5
e1	0	0	1	1	0

OPTIMAL

At the end of phase 1, artificial variable columns are deleted if they are all non-basic.

If an artificial variable is basic at zero level, it transfers to phase 2 as a part of the starting solution.

In such a case, that artificial variable should be forced to leave basis. It should be selected as the leaving variable even if its pivot column coefficient is negative or it is not the MR.

Special Cases in Simplex

Degeneracy
Alternative optima
Unbounded solution
Infeasibility

Degeneracy

If a tie occurs for MR, in the next iteration at least one basic variable will be zero. Such a solution is said to be degenerate

→ There is at least one redundant constraint
→ There could be cycling in simplex (However, its probability is small)

- There are methods to prevent cycling but very time consuming. Thus, usually it is not implemented.

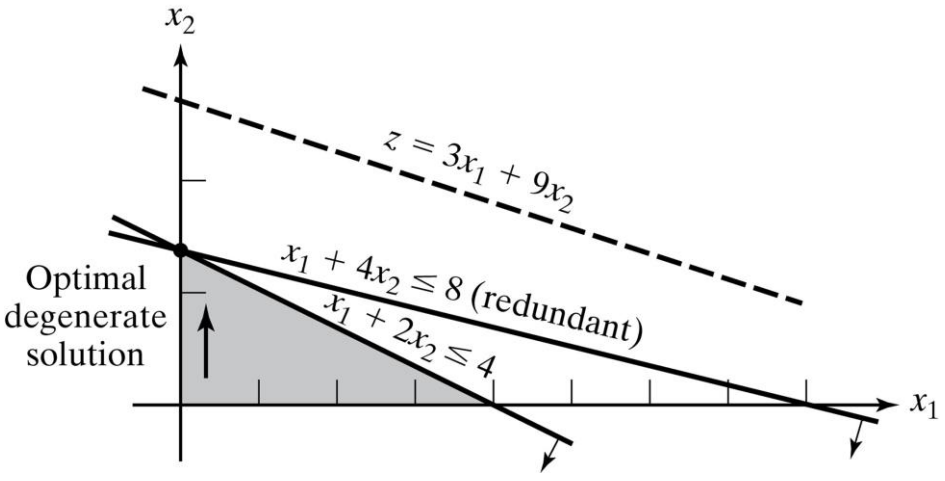
max $z = 3x_1 + 9x_2$

$x_1 + 4x_2 \le 8$

$x_1 + 2x_2 \le 4$

$x_1, x_2 \ge 0$

Basic	x1	x2	x3	x4	Solution
z	-3	-9	0	0	0
x3	1	4	1	0	8
x4	1	2	0	1	4
z	-3/4	0	9/4	0	18
x2	1/4	1	1/4	0	2
x4	1/2	0	-1/2	1	0
z	0	0	3/2	3/2	18
x2	0	1	1/2	-1/2	2
x1	1	0	-1	2	0

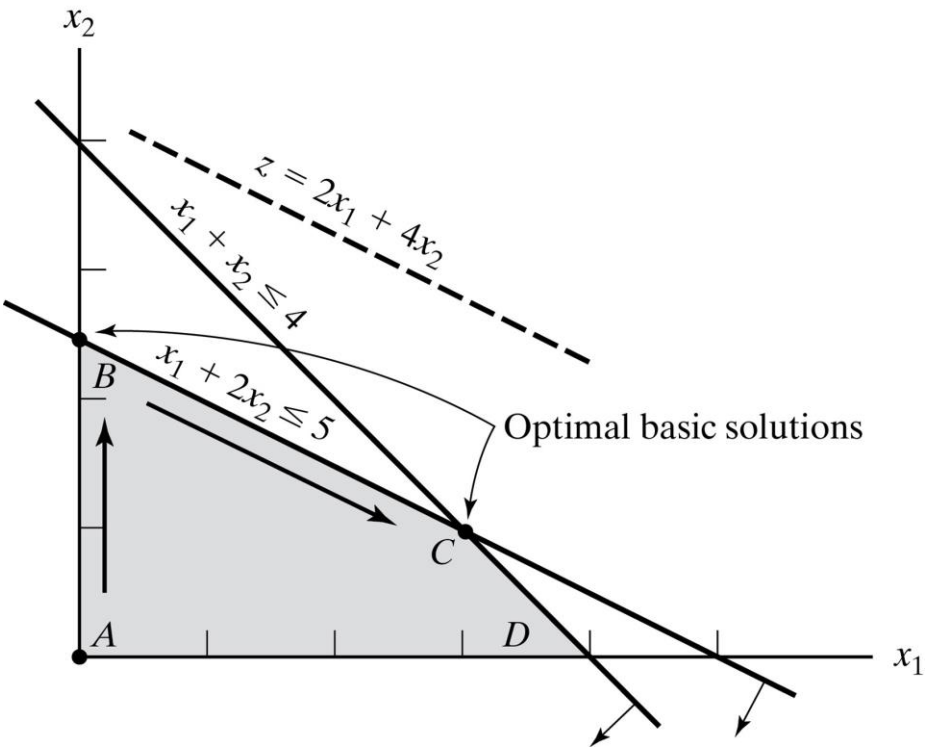


Alternative Optima

If the objective function lines are parallel to a binding constraint then, there are infinite numbers of alternative solutions with the same objective value.
Alternative optima can be detected in simplex if the z-row coefficient of a non-basic variable is zero. That means it can enter the basis without changing the objective value.

$$\begin{aligned} \max \quad & z = 2x_1 + 4x_2 \\ & x_1 + 2x_2 \le 5 \\ & x_1 + x_2 \le 4 \\ & x_1, x_2 \ge 0 \end{aligned}$$

Basic	x1	x2	x3	x4	Solution
z	-2	-4	0	0	0
x3	1	2	1	0	5
x4	1	1	0	1	4
z	0	0	2	0	10
x2	1/2	1	1/2	0	5/2
x4	1/2	0	-1/2	1	3/2
z	0	0	2	0	10
x2	0	1	1	-1	1
x1	1	0	-1	2	3

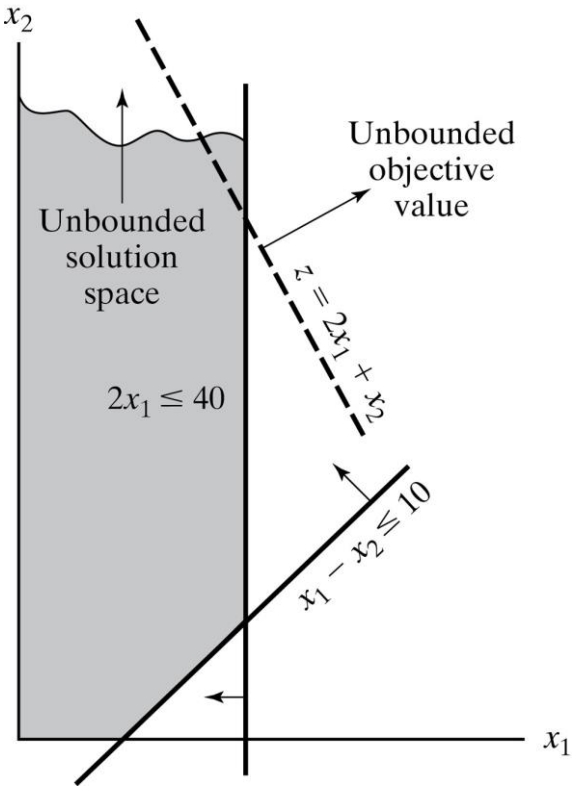


Unboundedness

$$\begin{aligned} \max \quad & z = 2x_1 + x_2 \\ & x_1 - x_2 \le 10 \\ & 2x_1 \le 40 \\ & x_1, x_2 \ge 0 \end{aligned}$$

Basic	x1	x2	x3	x4	solution
z	-2	-1	0	0	0
x3	1	-1	1	0	10
x4	2	0	0	1	40

- If the constant coefficients of any non-basic variable are all non-positive then, the feasible region is unbounded.
- If in addition the z-row coefficient of that variable is negative (in max. problem) or positive (in min. problem) then, the objective value is unbounded too.



Infeasibility

If the constraints are inconsistent then, there is no feasible solution of the problem.
Infeasibility can be detected if any artificial variable assumes a positive value at the final iteration of simplex. (both M-method and two-phase method)